

Title of your report

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In these guidelines we provide a more detailed idea of what is expected of your experiments and of the report. For each section there is a rough indication of how much text is expected and which figures. Depending on your results certain parts may require more or less text or more or fewer figures. The main goal of your experiments is to compare different initialization methods for Lloyd's algorithm for k-means.

1 Introduction

K-means clustering is an unsupervised learning method that partitions a dataset into k clusters. Lloyd's algorithm is a commonly used method for solving the k-means problem by repeatedly assigning points to the nearest centroid and then updating the centroids as the mean of the assigned points. Although the algorithm is simple and usually converges quickly, its final result can depend strongly on the initialization of the centroids.

This experiment compares three initialization methods for Lloyd's algorithm: random initialization, k-means++ initialization, and Gonzalez initialization. The experiment uses S&P 500 stock data, where each stock is represented by its yearly return and annualized volatility. This creates a two-dimensional risk-return dataset. This dataset is interesting because financial markets often contain outlier stocks with unusually high returns or volatility. Such outliers can strongly influence centroid initialization. In particular, Gonzalez initialization tends to select far-away points, which may cause it to react differently to outliers compared to random or k-means++ initialization.

How do outlier stocks affect the performance of random, k-means++, and Gonzalez initialization when clustering S&P 500 stocks by return and volatility?

2 Data sets and experiments

2.1 The dataset

The dataset used in this experiment consists of S&P 500 stock data from Kaggle [1], filtered to 2025. Each data point represents one stock. Instead of clustering the raw daily stock prices, every stock is represented by two summary features: annual return and annualized volatility. This creates a two-dimensional risk-return dataset, where the x-coordinate represents annual return and the y-coordinate represents annualized volatility.

The annual return is calculated from the stock price at the start and end of the measured period. The volatility is calculated as the standard deviation of the daily returns, multiplied

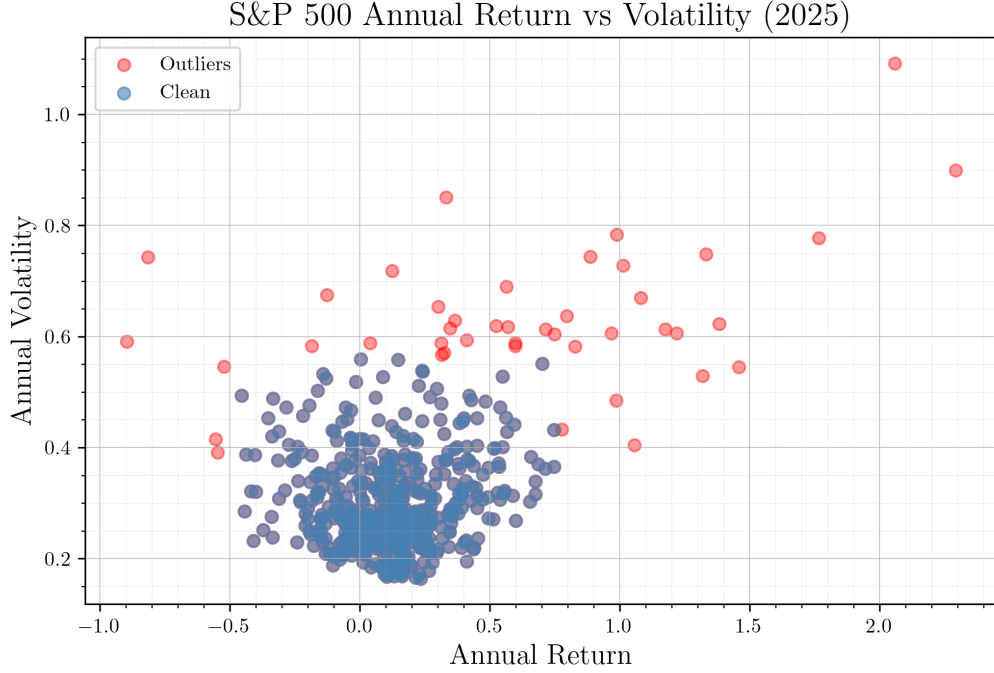


Figure 1: Outliers and cleaned dataset comparison

by $\sqrt{N}$, where N is the number of trading days in the measured period. The final dataset contains 503 stocks. Since financial data can contain stocks with unusually high returns or unusually high volatility, two versions of the dataset are used: the full dataset and a cleaned dataset where outliers are removed using the IQR method. A stock is considered an outlier if its annual return or annualized volatility lies outside the interval

$$[Q_1 - 1.5 \cdot IQR, Q_3 + 1.5 \cdot IQR].$$

2.2 The experiments

The experiment compares three initialization methods for Lloyd’s k-means algorithm: random initialization, k-means++ initialization, and Gonzalez initialization. In all three cases, only the initialization method changes. The same Lloyd method is then repeated: points are assigned to their nearest centroid using Euclidean distance, and centroids are updated as the mean of their assigned points until convergence.

Before clustering, the features are standardized so that annual return and annualized volatility contribute equally to the Euclidean distance calculation. The experiment is performed on both the full dataset and the outlier-removed dataset. This allows us to test whether outlier stocks affect the initialization methods differently. The experiments are run for different values of k , where we have selected the following values: $k = 3$, $k = 5$, and $k = 8$.

The methods are evaluated using inertia, silhouette score, number of Lloyd iterations, and runtime. Inertia measures the compactness of the clusters, while silhouette score measures how clearly separated the clusters are. The number of iterations and runtime are used to

compare convergence behavior and computational efficiency. Since random initialization and k-means++ use randomness, the experiments are repeated multiple times and average results are reported.

3 Experimental evaluation

[2-3 paragraphs with figures]

A presentation of the results of your experiments as well as a brief discussion of them. The presentation would generally be in the form of a graph or table. Be sure to focus on the interesting parts. If in most of your experiments the initialization methods you are comparing are doing exactly the same, then focus on the ones where they are not and just state that the others show the exact same results. In the discussion you consider what the results imply and discuss any potential irregularities or unexpected results.

Depending on your research question you can focus on several of the following measures.

- Convergence rate of the algorithm: How many iterations of Lloyd’s algorithm were needed before finishing?
- Cluster quality: What is the cost of the resulting clusterings?
- Actual running time: How long do the initialization and the rest of the algorithm take?

Note that for initialization methods that use randomness, you may need to run the same experiment multiple times and use an average value to obtain a reliable value for the convergence rate or the score. You should of course then also take this into account if you are comparing efficiency or runtime.

4 Concluding remarks

[1-2 paragraph]

A summary of the main conclusions of the experiments. Do your results help answer part of your initial research question?

References

- [1] DarkMatterNet. S&p 500 stocks: 25 years of data (updated daily). <https://www.kaggle.com/datasets/darkmatternet/s-and-p-500-stocks-25-years-of-data-updated-daily>, 2024. Accessed: 2026-05-25.